

# Melanie Fernández

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37-14 91st Street, Jackson Heights, NY 11372

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Computer science graduate with experience in software engineering, machine learning, data analysis, and technical research. Skilled in building reliable applications, developing data pipelines, and translating complex ideas into practical, reproducible solutions. Adaptable across new technologies and problem domains, with a strong foundation in problem-solving, experimentation, and end-to-end project development.

## TECHNICAL SKILLS

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**Languages:** Python, JavaScript, SQL, Java, C#, C++, R, HTML/CSS, Dart

**Backend/APIs:** Django, Django REST Framework, FastAPI, Flask, RESTful API Design, JWT Authentication, Flutter

**Frontend:** JavaScript, HTML/CSS, React.js, D3.js

**Databases:** SQL, MySQL, SQLite, PostgreSQL, Data Modeling, Query Writing

**ML/AI:** PyTorch, TensorFlow, scikit-learn, XGBoost, NumPy, pandas, JAX

**Dev Tools:** Git, Linux/Bash, VS Code, Vim/Neovim, Agile Collaboration, Debugging

## EDUCATION

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**B.A. Computer Science, Minor in Sociology** | Connecticut College | New London, CT

*Aug 2022 – May 2026*

- Focus: Machine Learning and Artificial Intelligence
- Relevant coursework: Data Structures, Algorithms, Computational Intelligence, Database Systems, Computer Networks, Data Visualization, Computer Organization
- Science Leaders Program — honors cohort integrating scientific and social research

## EXPERIENCE

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**Independent Software Developer** | Mercury Mental Healthcare | Freelance / Remote

*May 2026 – Present*

- Developing a browser-based mental health survey platform from client requirements through deployment, rapidly iterating on user experience, data collection workflows, and reporting functionality.
- Debugging frontend and data-flow issues across browser-based deployment environments to ensure stable survey completion and reliable response storage.
- Collaborating directly with stakeholders to refine requirements, prototype features, and deploy updates in response to user feedback.
- Designing secure and user-friendly interfaces optimized for iPad/web deployment while considering HIPAA-compliant data handling practices.

**ML Research Engineer** | Autonomous Agent Learning Lab, Connecticut College | New London, CT

*Jun 2023 – May 2026*

- Engineered Python data pipelines to ingest, clean, and transform large structured datasets, reducing preprocessing time by ~35% across multiple ML experiments.
- Designed and trained CNN models on board-state representations for the strategy game Shogi, achieving results published at an international peer-reviewed conference.
- Improved experimental reproducibility by standardizing training workflows and model checkpointing, enabling faster iteration cycles across the research team.

**Teaching Assistant — Algorithms & Computational Intelligence** | Connecticut College | New London, CT

*Sep 2025 – May 2026*

- Supported undergraduate students in algorithmic reasoning, computational problem-solving, and machine learning concepts through one-on-one guidance and collaborative instruction.
- Worked closely with faculty to reinforce classroom material and improve student understanding across varying technical backgrounds.

**ML Research Fellow** | SZTAKI MILAB / AIT Budapest | Budapest, Hungary

*Aug 2024 – Dec 2024*

- Built end-to-end Python pipelines for cleaning, validation, and transformation of biomedical datasets at scale.
- Implemented KNN imputation and regression models to recover missing values, improving dataset completeness from 62% to 95% and enabling downstream model training.
- Collaborated with computational biology researchers to evaluate model reliability on aging biomarker data, translating domain requirements into ML solutions.

**Data Analyst** | Connecticut College | New London, CT

Sep 2022 – Dec 2022

- Cleaned, validated, and standardized alumni datasets using SQL and Excel to support institutional reporting and improve long-term data reusability.

## PROJECTS & PUBLICATIONS

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### SupportHub — Full-Stack Ticketing System

2024

- Built a full-stack support platform using Django REST Framework and JWT authentication, enabling role-based workflows for customers, staff, and managers.
- Designed RESTful APIs and frontend interfaces supporting ticket creation, assignment, tracking, and resolution across multiple user roles.

### Trader Joe's Retail Database System

2026

- Designed and implemented a normalized MySQL database modeling retail operations across physical and online stores, including inventory, transactions, customers, vendors, and restocking workflows.
- Built command-line workflows supporting customer checkout, cashier operations, inventory updates, and restocking simulations using SQL and Bash.

### Interactive Star Data Visualization

2024

- Built an interactive astronomical data explorer using D3.js with dynamic filtering, responsive visualizations, and real-time analysis of large star datasets.

### Cross-game position-value transfer in chess and shogi with aligned per-game encoders

2026

- Engineered a CORAL-aligned deep learning model that significantly improved cross-game value prediction between chess and shogi over shared-trunk and handcrafted feature baselines.

*Published: Fernández, O'Connor, & Parker, AIIDE 2026 (Belo Horizonte, Brazil)*

### MAGE: MAP-Elites for Alpha Generation

2026

- Applied MAP-Elites and genetic programming to generate diverse and robust alpha trading factors on S&P 500 market data for quantitative finance research.

*Pending Publication (CIEr): O'Connor & Fernández (2026)*

### Online Shogi Match Prediction — Deep CNN

2023 – 2025

- Built and trained a deep CNN to predict online match outcomes from board-state representations, iterating across multiple model architectures to improve predictive accuracy.
- Extended the model with improved architectures and deeper analysis, accepted as a post-publication extension in *Computational Intelligence*, Springer 2025.
- Nominated for Best Student Paper at IJCCI 2024 in Porto, Portugal.

*Published: O'Connor & Fernández, IJCCI 2024 (Porto) · Extended: Computational Intelligence, Springer 2025*

### Simulation of Speciation and Complex Trait Evolution

2025 – 2026

- Contributed to a multi-species artificial environment simulation modeling evolutionary speciation and complex trait development across agents.

*Published: Parker, O'Connor, Nash & Fernández, ICAART 2026 (Marbella, Spain)*